

Experience the Virtual World with OpenVZ

OpenVZ is container based, open source, virtualisation technology for Linux. It can be used to install multiple, fully isolated Linux containers easily on a single host machine. Apart from its low overhead, OpenVZ has many advantages over other virtualisation systems like KVM.



OpenVZ, as most of us know, has taken the field of computer virtualisation by storm. In this article, I will discuss its various aspects and how to implement it with your infrastructure, in order to create virtual machines within minutes. These machines will be isolated from one another and only the host machine will have full control over them. I will walk you through the steps of getting OpenVZ up and running on a CentOS 6 server, along with a few configuration tweaks.

What is OpenVZ?

OpenVZ is open source virtualisation technology with which you can create one or more self-contained operating systems, isolated from one another, but which behave just like standalone systems or servers. For the more technically oriented, it is a framework that is used to divide or share the host server's resources among self-contained virtual containers, which share the host machine's patched kernel in order to run.

How is OpenVZ different from KVM?

KVM stands for *Kernel based Virtual Machine*, also referred to as full virtualisation. It is used to create totally isolated virtual machines, which contain their own kernels and own

dedicated resources such as disks, graphics cards, network cards, etc. Virtual KVM servers behave like completely isolated environments.

OpenVZ is container based virtualisation, in which the resources of the host machine are shared among a number of virtual containers.

This is the main reason why OpenVZ can create only Linux based virtual machines (VMs) whereas KVM can create virtual servers that are based on Linux, Windows, BSD, Solaris, etc, because they are self-contained, have their own resources, and do not need to rely on the host machine. Although KVM also must be installed on a Linux box, it can have other OSs as guest machines, as mentioned above.

Because of this full virtualisation, KVM generally has much more overhead than OpenVZ containers, which makes OpenVZ pretty fast compared to KVM. All in all, choosing between OpenVZ and KVM should depend on what you are going to run on the virtual server. Given below are some comparisons that you could consider when making a choice.

1. *You want the server to host your PHP/MySQL application:*

Go with OpenVZ, as your application must be fast enough to cater to the requests of your clients - an OpenVZ based